

ELEVATE LABS CYBER SECURITY INTERNSHIP

Task-1 Local Network Port Scanning Using Nmap

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Domain: Cyber Security

Organization: Elevate Labs

Tool Used: Nmap, Kali Linux

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Cyber Security Internship - Task 1: Local Network Port Scanning

Step-by-Step Implementation

Step 1: Finding Local IP Address & Network Range

Before running a scan, we checked our system's network configuration and active IP address using the `ip a` and `ifconfig` commands on Kali Linux.

- **Screenshot 1: Checking IP configuration (`ip a`)**

```
(mansiraviya@kali)-[~]
└─$ sudo su
[sudo] password for mansiraviya:
(root@kali)-[/home/mansiraviya]
└─$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:db:80:02 brd ff:ff:ff:ff:ff:ff
    inet 10.207.70.140/24 brd 10.207.70.255 scope global dynamic noprefixroute eth0
        valid_lft 3508sec preferred_lft 3508sec
    inet6 fe80::a00:27ff:fedb:8002/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
```

- *Observation:* Our local interface `eth0` has the IP address `10.207.70.140`, which means our target local network subnet range is `10.207.70.0/24`.

- **Screenshot 2: Verifying with `ifconfig`**

```
(root@kali)-[/home/mansiraviya]
└─$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.207.70.140 netmask 255.255.255.0 broadcast 10.207.70.255
    inet6 fe80::a00:27ff:fedb:8002 prefixlen 64 scopeid 0x20<link>
    ether 08:00:27:db:80:02 txqueuelen 1000 (Ethernet)
    RX packets 3918 bytes 4877845 (4.6 MiB)
    RX errors 419 dropped 0 overruns 0 frame 419
    TX packets 2806 bytes 240413 (234.7 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 8 bytes 480 (480.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 8 bytes 480 (480.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

- *Observation:* Confirmed the subnet mask (`255.255.255.0`) and broadcast address (`10.207.70.255`).

Step 2: Running Nmap TCP SYN Scan

We executed a stealth TCP SYN scan (-sS) across our local subnet range (10.207.70.0/24) to detect live hosts and open ports without completing the full 3-way handshake.

- **Command Used:**

Bash

```
sudo nmap -sS 10.207.70.0/24
```

- **Nmap Scan Output**

```
(root@kali)-[/home/mansiraviya]
└─$ sudo nmap -sS 10.207.70.0/24
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-27 16:32 +0530
Nmap scan report for 10.207.70.175
Host is up (0.0033s latency).
Not shown: 999 closed tcp ports (reset)
PORT      STATE SERVICE
53/tcp    open  domain
MAC Address: 1E:6F:A9:03:88:D7 (Unknown)

Nmap scan report for 10.207.70.189
Host is up (0.00058s latency).
Not shown: 999 filtered tcp ports (no-response)
PORT      STATE SERVICE
7070/tcp  open  realserver
MAC Address: 28:D0:EA:5B:D0:B0 (Intel Corporate)

Nmap scan report for 10.207.70.140
Host is up (0.000090s latency).
Not shown: 999 filtered tcp ports (no-response)
PORT      STATE SERVICE
22/tcp    open  ssh

Nmap done: 256 IP addresses (3 hosts up) scanned in 18.28 seconds
```

Scan Results & Observations

Out of 256 IP addresses scanned, **3 live hosts** were discovered within 18.28 seconds:

1. **Host 1: 10.207.70.175**
 - **Open Port:** 53/tcp
 - **Service:** Domain (DNS)
 - **MAC Address:** 1E:6F:A9:03:88:D7 (Unknown)
2. **Host 2: 10.207.70.189**
 - **Open Port:** 7070/tcp
 - **Service:** Realserver
 - **MAC Address:** 28:D0:EA:5B:D0:B0 (Intel Corporate)
3. **Host 3: 10.207.70.140 (Local Machine)**
 - **Open Port:** 22/tcp
 - **Service:** SSH (Secure Shell)

Conclusion & Security Insights

- **Network Exposure:** Open ports like SSH (22) and DNS (53) are common in network environments but require proper configuration (such as key-based authentication or firewall restrictions) to prevent unauthorized access.
- **Reconnaissance Skills Learned:** Successfully utilized Nmap for identifying live hosts, determining port states, and recognizing service exposure risks on a local network